Prodigy Anesthesia Quick Review

Hypertension

What is 'pre-hypertension'?	Prehypertension is defined as a systolic blood pressure between 120-139 or a diastolic blood pressure between 80-89. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 104.*
How common is hypertension in the United States?	Approximately 70 million Americans are diagnosed with hypertension. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 5418.*
How is the diagnosis of hypertension made in the adult patient?	Hypertension is diagnosed when the blood pressure is determined to be at least 140/90 on two successive occasions that are at least 1-2 weeks apart. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 87.*
What is the difference between essential hypertension and remedial (secondary) hypertension?	Secondary hypertension is an increase in blood pressure due to a cause that can be identified and cured such as pheochromocytoma, renal artery stenosis, coarctation of the aorta, Conn's syndrome, or Cushing disease. Essential hypertension is an increased blood pressure for which there is no identifiable cause and is diagnosed based on the exclusion of causes such as those listed above. Approximately 95% of patients with hypertension are diagnosed with essential hypertension. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 491.*
What is the most common cause of secondary hypertension?	Renal artery stenosis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 105.*
What lifestyle modifications have been shown to be effective in the treatment of hypertension?	Smoking cessation, weight loss, exercise, maintenance of normal calcium and potassium levels, and limiting sodium intake. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 106-107.*
What are the positive effects of sodium restriction?	Decreasing sodium intake has been shown to reduce blood pressure, minimize hypokalemia due to diuretic use, protect from osteoporosis, and enhance ventricular contractility as a result of decreased calcium excretion. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 107.*

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How is the determination made to institute lifestyle modification changes versus placing a patient on antihypertensive drugs?	If the patient exhibits any signs of end-organ damage due to hypertension such as renal disease, coronary artery disease, cerebrovascular disease, retinopathy, etc,) the patient will be placed on antihypertensive therapy as soon as possible. If no evidence of end-organ damage is present and the patient is otherwise asymptomatic, lifestyle modification therapy is often effective. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 106-107.*
What concomitant disorders are often seen with hypertension?	Hypertension is often seen with diabetes mellitus, insulin resistance, dyslipidemia, and obesity. Approximately 40% of patients with hypertension have elevated cholesterol levels. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 104-105.*
How is the degree of hypertension related to cardiovascular risk?	For every increase of 20/10 mmHg over a 115/75, the risk of cardiovascular disease doubles. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 491.*
How does the institution of antihypertensive medication in patients with essential hypertension affect their cardiovascular risk?	The implementation of antihypertensive therapy reduces the risk of cardiovascular complications by 25% and reduces the risk of stroke by 38%. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 491.*
How is alcohol ingestion related to hypertension and cardiovascular risk?	Alcohol ingestion is associated with an increase in blood pressure, but the moderate consumption of alcohol has been shown to decrease overall cardiovascular risk. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 105.*
How does chronic hypertension affect the left ventricular oxygen supply and demand?	As the left ventricle undergoes compensatory concentric hypertrophy, the increased myocardial mass has a higher oxygen demand while the contraction of the hypertrophied muscle can reduce subendocardial oxygen supply, placing the myocardium at greater risk for ischemia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 492.*
How does the left ventricle respond to the increased demands made on it by hypertension?	Faced with the increased afterload associated with chronic hypertension, the left ventricle increases its myocardial mass (concentric hypertrophy). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 492.*
What is the relationship between hypertension and atherosclerotic disease?	Hypertension accelerates the development of atherosclerotic changes in the arterial vessels. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 491.*

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What is the standard goal of therapy in the medical management of hypertension? How is this influenced by a concomitant diagnosis of diabetes mellitus or renal disease?	The standard goal is the reduction of the blood pressure to less than 140/90. In patients with diabetes mellitus or renal disease, the goal is to reduce the blood pressure to less than 130/80 mmHg. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 105.*
How is obstructive sleep apnea linked to hypertension?	Obstructive sleep apnea results in paroxysmal hypoxemia, arousal, and activation of the sympathetic nervous system with resulting increases in blood pressure. Approximately 30% of patients with OSA are also diagnosed with hypertension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 105.*
How can hypertension lead to congestive heart failure?	The chronic increase in afterload can result in cardiomyopathy and left ventricular failure with resultant congestive heart failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 491.*
How is the renal juxtoglomerular apparatus implicated in the pathophysiology of hypertension?	In the face of chronic vasoconstriction, the response of the juxtaglomerular apparatus is to increase intravascular volume. It releases renin. The end result of renin release is the conversion of angiotensin I to angiotensin II. Angiotensin II is the primary stimulus for the release of aldosterone by the adrenal cortex. Aldosterone results in increased sodium and water retention. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 492.*
Within what range should you maintain blood pressure during the anesthetic for a patient with chronic hypertension?	The goal is maintenance of the mean arterial blood pressure within 20% of their normal mean pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 493.*
What medications may be used to treat intraoperative hypertension in the chronically hypertensive patient?	Volatile anesthetics and opioids may be used to control blood pressure, as may beta-blockers, ACE inhibitors, nitroprusside, alpha-2 agonists such as clonidine, alpha-1 blockers such as droperidol, and calcium-channel blockers. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 494.*
What are methods that can be employed to reduce the exaggerated blood pressure response to laryngoscopy seen in many hypertensive patients?	Continuing their regularly scheduled antihypertensive medication, achieving a deeper level of anesthesia than typical for non-hypertensive patients prior to laryngoscopy, pre-treating with lidocaine intravenously or administration of 2-3 mcg/kg of fentanyl, and reducing the duration of laryngoscopy to less than 15 seconds. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 493.*

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What is the most effective means of preventing the postinduction hypotension commonly seen in hypertensive patients?	Because the chronically hypertensive patient is volume contracted, adequate hydration prior to induction will usually attenuate the drop in blood pressure seen after induction of anesthesia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 493.*
How can antihypertensive medical therapy affect homeostatic compensatory mechanisms and how can this affect the anesthetic management of the patient?	Antihypertensives can block the body's ability to compensate for changes in sympathetic tone. As a result, the normal reflexes such as increased heart rate or vasoconstriction that may normally occur during induction of anesthesia or in response to blood loss may not occur, making the anesthetic management of the patient more difficult. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 492.*
How rapidly should the blood pressure be decreased in patients with hypertensive emergency and why?	The goal is reduction of the diastolic pressure by about 20% over the first 60 minutes and then more gradually after that. Reducing the diastolic pressure too rapidly can result in coronary or cerebral ischemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 109.*
What are the most commonly used methods of treating intraoperative hypotension in the chronically hypertensive patient?	Prolonged hypotension in the hypertensive patient is associated with poor outcomes and must be treated aggressively. If reducing the amount of volatile agent used and ensuring adequate volume expansion do not treat the problem rapidly, then fast-acting vasopressors such as phenylephrine or ephedrine may be necessary until the cause of the hypotension can be identified. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 494.*
At what blood pressure does encephalopathy usually appear? What is the exception to this rule?	Encephalopathy usually does not present until the diastolic blood pressure reaches 150 mmHg. The exception is parturients who may exhibit signs of encephalopathy with diastolic pressures at 100 mmHg. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 109.*
Why should nitroprusside be used with caution in the hypertensive patient exhibiting encephalopathy?	Nitroprusside's use can be complicated by lactic acidosis and cyanide toxicity. Nicardipine (which offers both cardiac and cerebral protection) may be a more suitable option. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 110.*

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What is the difference between hypertensive emergency and hypertensive urgency?	Hypertensive emergency is defined as hypertension with evidence of end-organ damage such as myocardial ischemia, dissecting aortic aneurysm, renal insufficiency, pulmonary edema, encephalopathy, eclampsia, or intracerebral hemorrhage. Hypertension without signs of end-organ damage is termed 'hypertensive urgency'. These patients often present with hypertension and symptoms such as headache, epistaxis, or anxiety. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 109.*
How is the diagnosis of hypertensive emergency made in parturients?	Even if no overt signs of end-organ damage are noted, a diastolic pressure higher than 109 mmHg in a parturient is designated as a hypertensive emergency and immediate treatment is warranted. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 109.*
How is hypertension due to primary aldosteronism (Conn's syndrome) treated differently in males and females and why?	Primary aldosteronism is treated with spironolactone in women. Amiloride is used as the treatment of choice in males as spironolactone can result in gynecomastia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 109.*

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